

FDN 1202

Introduction to Computers

Classification of Computers

What is a Computer ?

- ❑ The computer is an Electronic Device which
 - accepts input data
 - processing according to their instruction
 - and gives output result.



Computer Classification Factors

- ❑ Computers may be classified based on the following:
 - Type of Data Handled.
 - Size and Processing Powers.
 - Function or Purpose.

Classifications of Computers According to Type of Data Handled

```
graph TD; A[Classifications of Computers According to Type of Data Handled] --> B[Digital Computers]; A --> C[Analog Computers]; A --> D[Hybrid Computers];
```

Digital
Computers

Analog
Computers

Hybrid
Computers

Digital Computers

- Operates with information represented in numerical or digital form.
- Process data (including text, sound, graphics and video) into a digital value, in 0s and 1s.
- In digital computers analog quantities must be converted into digital quantity before processing. In this case output will also be digital.
- If analog output is desired, the digital output has to be converted into analog quantity.
- The components, which perform these conversions, are the essential parts or peripherals of the digital computer.

- Digital computers can give the results with more accuracy.
- Examples:
 - Desktop
 - Laptop
 - Cash Registers
 - Smart phones



Analog Computers

- Operates on data in the form of continuously variable physical quantities such as electrical potential, fluid pressure, mechanical motion etc.
- These computers do not deal directly with the numbers.
- They measure continuous physical magnitudes (e.g. temperature, pressure and voltage), which are analogous to the numbers under consideration.
 - ❑ As an example petrol pump may have an analog computer that converts the flow of pumped petrol into two measurements: the quantity of petrol and the price of that quantity.

- Analog computers
 - ❑ Use for scientific and engineering purposes.
 - ❑ They give approximate results.
 - ❑ They are very fast in operation.
 - ❑ Easy to get graphical results directly.
 - ❑ The accuracy of analog computers is less.
 - ❑ All the analog computers are special purpose computers.

- Examples:
 - ☐ Speedometer of a car.
 - ☐ Seismometer.
 - ☐ Gasoline Pump.



Hybrid Computers

- Incorporated the measuring feature of an analog computer and counting feature of a digital computer.
 - ❑ For computational purpose – use the analog components.
 - ❑ For the storage of intermediate results – digital memories are used.
- These computers have the ability to accept the input data either analog or digital form.
- Used in scientific applications, various fields of engineering and industrial control process.



Classifications of Computers According to Function or

```
graph TD; A[Classifications of Computers According to Function or] --> B[Special Purpose Computers]; A --> C[General Purpose Computers];
```

**Special Purpose
Computers**

**General Purpose
Computers**

Special Purpose Computers

- A special purpose computer is designed only to meet the requirements of a particular task or application.
- The instruction needed to perform a particular task are permanently stored into the internal memory, so that it can perform the given task on a single command.
- They cannot be used for other applications unless their circuits are redesigned.
- Can provide the result very quickly and efficiently.

- Special Purpose computers are used for airline reservations, satellite tracing , air traffic control and etc.
- Examples:
 - ☐ ATM Machine
 - ☐ Washing Machine
 - ☐ Cash Registers



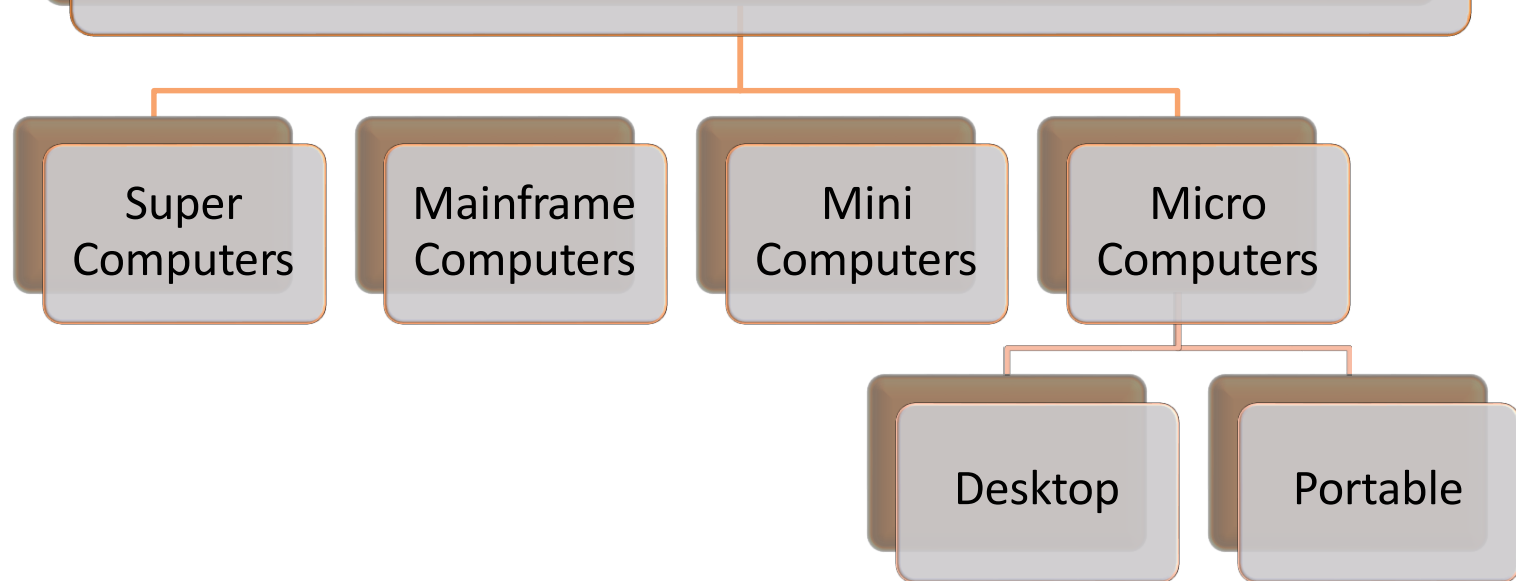
General Purpose Computers

- A General Purpose Computers are designed to meet the needs of many different applications.
- In these computers, the instructions needed to perform a particular task are wired permanently into the internal memory. When one job is over, instructions for another job can be loaded into the internal memory for processing.
- They are versatile, and generally lack in speed and efficiency.

- The computers used in schools and homes are general purpose computers.
- Examples:
 - ❑ Desktop Computers.
 - ❑ Laptop.



Classifications of Computers According to Size and Processing Power



Super Computers

- Super computers have extremely large storage capacity and computing speed which are many times faster than other computers.
- It basically contains a number of CPUs that operate in parallel to make it faster.
- A super computer is measured in terms of tens of millions Instructions per second, an operation is made up of numerous instructions.
- These are very expensive and very large in size.
 - ❑ Accommodate in large air-conditioned rooms; some super computers can span an entire building.

- These are special purpose computers which designed to work on a single problem at a time.
- Super computers are used for highly calculation intensive tasks such as problems including
 - ❑ Quantum physics
 - ❑ Weather forecasting
 - ❑ Climate research
 - ❑ Molecular modeling
 - ❑ Physical simulations
 - ❑ Aircraft and spacecraft designing
 - ❑ Encrypt and decrypt sensitive intelligence information
 - ❑ Nuclear weapon testing



- Example:
 - IBM Deep Blue
 - Fujitsu's K Computer in Japan
 - IBM's Mira in USA
 - IBM's Super MUC in Germany



Deep Blue computer play a chess game against Garry Kasparov in 1997, it is an IBM supercomputer that uses scalable parallel processing to solve complex problems. Deep Blue uses 256 processors working together to calculate between 50 and 100 billion chess moves in under three minutes.

Mainframe Computers

- A mainframe computer is a very large computer capable of handling and processing very large amounts of data quickly.
- These are the second largest (in capability and size) of the computer family.
- Mainframe computers are used in large organizations such as insurance companies and banks, where many people frequently need to use the same data.
- Mainframe computer allows its user to maintain large information storage at a centralized location and is able to access and process this data from different computers located at different locations.

- In a traditional mainframe environment, each user access the mainframe's resources through a device called terminal.
- Two kinds of terminals are available.
 - Dumb terminal : it is simply an input/output device that function as a window into a computer located somewhere else.
 - Intelligent terminal : can perform some processing operations but usually does not have any storage.

- Mainframe computers are often housed alone in special rooms, away from their users.
- Examples:
 - IBM 370
 - S/390
 - VAX 8000
 - CDC 6600



Mini Computers

- A mini computer is a medium sized computer.
- Able to process and store less data than a mainframe, but more than a micro computer.
- Since the capabilities of the minicomputers somewhere between those of mainframe and microcomputers these are known as “midrange computers”
- These computers are usually designed to serve multiple users simultaneously.
 - It is capable of supporting 4 to about 200 simultaneous users.
- They are more expensive than micro computers.

- Examples:
 - Digital Alpha
 - Sun Ultra



Microcomputers

- A microcomputer is the smallest, low cost, single user and general purpose digital computer.
- The older PC started 8 bit processor with speed of 3.7 MB and current PC 64 bit processor with speed of 66 GB.
 - Examples:
 - IBM PC based on Pentium microprocessor and APPLE Macintosh.
- They consists of CPU, input unit, output unit, storage unit and the software.

- Although minicomputers are stand-alone machines they can be connected together to create a network of computers that can serve more than one user.
- Can be classified into two types.
 - Desktop
 - Portable
 - The difference is portable can be used while travelling where as desktop cannot be.

Portable Computers

- The different portable computers are
 - Laptop
 - Notebooks
 - Palmtop
 - Wearable Computers

Laptop

- They are portable and have all the features of a desktop computer but the size is smaller.
- They are expensive than desktop.
- The advantages of the laptop is that it is
 - Smaller in size
 - Can be carried anywhere
 - Has battery backup
 - Has all the functionality of the desktop
- Laptops can be placed on the lap while working (hence the name)



Notebook

- These are smaller notebooks optimized for weight and low cost and are designed for accessing web-based applications.
- Notebooks deliver the performance needed to enjoy popular activities such as streaming videos or music, emailing, web surfing or instant messaging.
- The word notebook was created as a blend of Internet and notebook.



Palmtop

- They are also called as Personal Digital Assistant (PDA).
- These computers are small in size and they can be held in hands.
- It is capable of doing word processing, spreadsheets and hand writing recognition, game playing and faxing.
- These computers are not as powerful as desktop computers.
- Examples:
 - 3com palmV

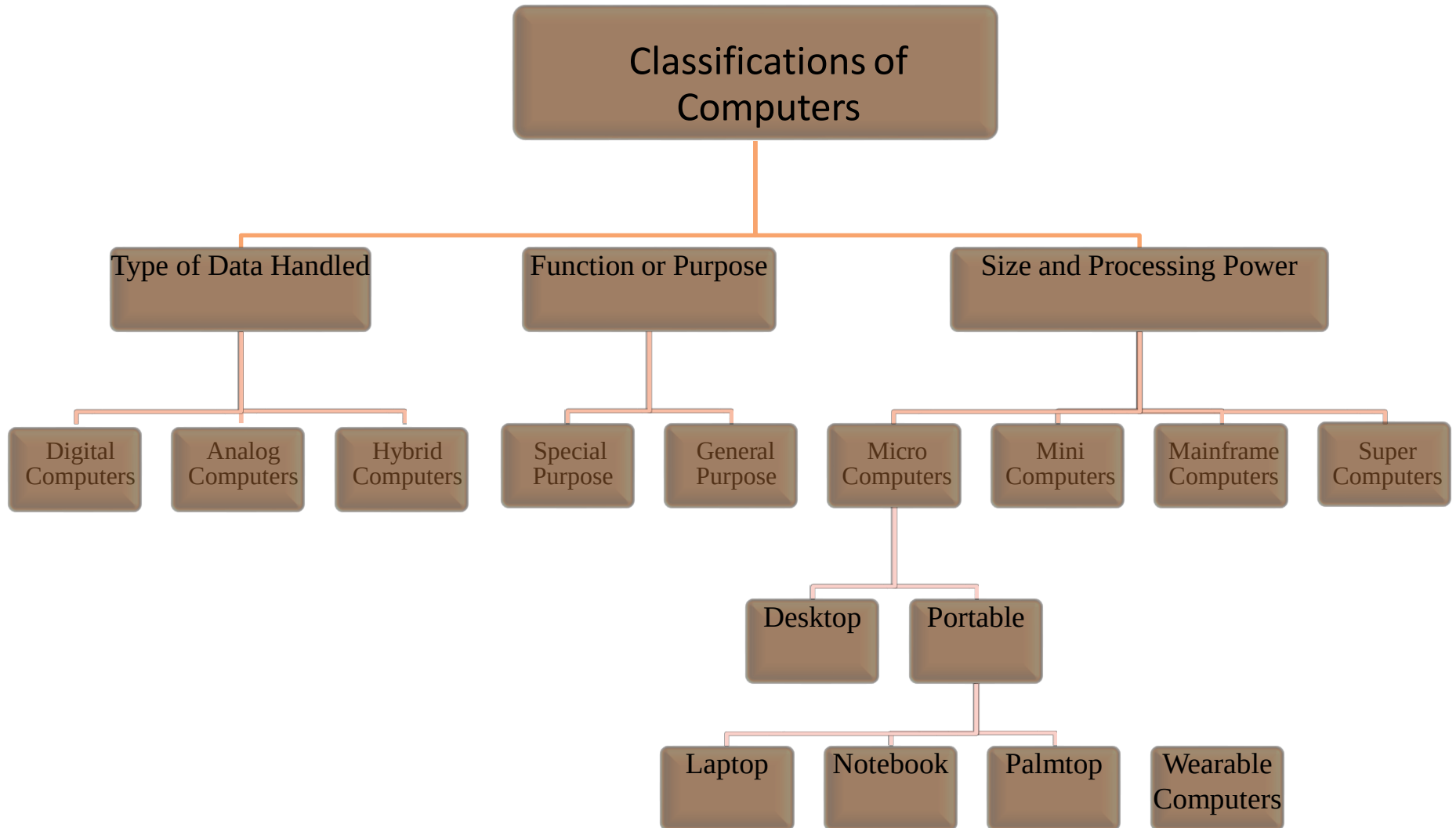


Wearable Computer

- The size of this computer is very small so that it can be worn on the body.
- It has smaller processing power.
- It is used in the field of medicine.
- Examples:
 - Pace maker to correct the heart beats.
 - Insulin meter to find the levels of insulin in the blood.



Summary



THANK YOU